ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) | OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: HARANYA.I

Roll No: 26CS246

Date:

Subject Name: Professional English Lab Integrated

DEPARTMENT OF HUMANITIES & SCIENCES | AFFILIATED TO ANNA UNIVERSITY | CHENNAI

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PROFESSIONAL SELF-INTRODUCTION

Haranya I

B.E. Computer Science and Engineering | St. Joseph's College of Engineering

Subject: Professional English

1. Oral Presentation Script

> *"Good morning, Sir. Thank you for giving me this opportunity to introduce myself. My name is **Haranya I**, and I am a first-year student pursuing a Bachelor of Engineering in Computer Science and Engineering at **St. Joseph's College of Engineering**. Technology has always fascinated me because a single idea, written as code, can become a tool that serves millions of people. My ambition is to become a skilled **Software Engineer and Developer**. I draw my inspiration from **Mr. Mark Zuckerberg**, the founder of Facebook, who showed that a determined developer with a clear vision can build a platform that connects the world. I am eager to strengthen my fundamentals, work on practical projects, and collaborate with peers who share my passion. Thank you, Sir."*

2. Personal and Academic Profile

Full Name	Haranya I
Subject	Professional English
Academic Status	First-Year Undergraduate Student (B.E., 1st Year)
Degree & Department	Bachelor of Engineering in Computer Science and Engineering (B.E. CSE)
Institution	St. Joseph's College of Engineering
Technical Interests	Software development, programming fundamentals, and problem solving
Career Aspiration	Software Engineer and Developer
Role Model	Mr. Mark Zuckerberg, Founder of Facebook

3. Academic Foundation in Computer Science

As a first-year student of Computer Science and Engineering at St. Joseph's College of Engineering, I regard this stage of my education as the time to build a solid and lasting foundation. My present curriculum introduces me to mathematical reasoning, logical thinking, and the basic principles of programming. These are the tools on which every advanced concept in computing is built, and I am committed to understanding them thoroughly rather than merely memorising them.

I do not limit my learning to lectures and textbooks. I make a deliberate effort to practise programming regularly, to attempt problems independently before seeking help, and to learn from every error I encounter in the laboratory. I believe that real competence comes from consistent practice, and that curiosity combined with discipline is the surest way to turn theoretical knowledge into dependable technical skill.

4. Long-Term Professional Career Goal

The Vision of Becoming a Software Engineer and Developer

My ultimate goal is to become an accomplished **Software Engineer and Developer**. To me, software development is far more than writing lines of code. It is the discipline of understanding a real problem, breaking it into manageable parts, and designing a solution that is efficient, reliable, and easy for people to use. I want to create applications that make everyday life simpler and that remain useful as technology changes.

My Planned Path of Growth

To achieve this vision, I have set out a clear direction for my years in engineering:

- **Strong fundamentals:** master programming languages, data structures, and algorithms so that I can write clean and efficient logic.
- **Practical development:** build small projects in web and application development to apply what I learn in class.
- **Modern technologies:** explore databases, version control, and cloud platforms that are widely used in the industry.
- **Professional skills:** develop communication, teamwork, and time management, since software is almost always built in collaboration.
- **Continuous learning:** keep up with new tools and practices, because technology never stops evolving.

5. Professional Inspiration

The Visionary Journey of Mr. Mark Zuckerberg

My greatest source of inspiration in the technology world is **Mr. Mark Zuckerberg**, the founder of Facebook, which is now part of Meta. He began building Facebook in 2004 while he was still a university student, writing the first version himself. What started as a small project among fellow students grew into a platform that connects billions of people across the world. His story shows me that a single idea, backed by programming skill and determination, can reach a global audience.

From a technical point of view, I admire his commitment to building and improving products continuously. Under his leadership, the company grew from a student project into one of the most influential technology organisations, and it has also contributed widely used open-source tools to the developer community. This culture of building, testing, and sharing is something I wish to adopt in my own work.

What I Learn from His Journey

- **Start with what you can build:** he began with code and a clear idea, which reminds me to start small and keep improving.
- **Persistence:** building a lasting product demands patience and the ability to keep going through difficulties.
- **Think at scale:** a developer should consider how a solution can serve many people, not just a few.
- **Learn and adapt:** successful technologists keep learning as the industry changes.

As an aspiring software engineer and developer, I hope to combine his spirit of innovation with humility, integrity, and a strong sense of responsibility towards the people who use what I create.

PERSONAL CAREER ROADMAP

SWOC Analysis

From First-Year Student to Skilled Software Engineer

Name	Haranya I
Institution	St. Joseph's College of Engineering
Department	Computer Science and Engineering (CSE), First Year
Subject	Professional English
Career Goal	Software Engineer and Developer, with a focus on cloud computing and cyber security

1. Career Vision

I am a first-year Computer Science and Engineering student, and my aim is to grow into a skilled **software engineer and developer** with strong knowledge of **cloud computing and cyber security**. This analysis uses the SWOC framework to examine my strengths, weaknesses, opportunities, and challenges honestly, and then turns them into a practical four-year plan.

2. SWOC at a Glance

S | Strengths

- Learns new concepts quickly
- Works well in a team
- Understands concepts clearly instead of memorising
- Knows C, C++, and Python

W | Weaknesses

- Needs better time management
- Finds speaking in front of people difficult
- Still at a beginner stage, with limited practical experience

O | Opportunities

- Growing demand for cloud computing and cyber security skills
- Free student learning programs and online courses
- Entry-level certifications, hackathons, and open-source projects
- College clubs, workshops, and technical events

C | Challenges

- Technology changes fast and needs constant learning
- No internship experience yet
- Growing competition for entry-level roles

3. Action Plan

Each weakness and challenge is paired with a specific step I will take to overcome it:

Time management	Plan each week in advance, set deadlines for every task, and review progress every Sunday.
Public speaking	Practise presentations in front of friends, volunteer for class talks and seminars, and join a technical or speaking club.
Limited experience	Build small projects in C++ and Python regularly and upload them to a public GitHub profile.
Fast-changing technology	Master core fundamentals first, then follow a fixed weekly time for learning new tools.
No internship yet	Apply for virtual internships, take part in hackathons, and contribute to open-source projects while waiting for a formal internship.

4. Four-Year Milestone Roadmap

Year	Primary Focus	Target Outcome
Year 1	C, C++, and Python fundamentals, problem solving, Linux basics, and cloud basics	Complete small projects; earn a foundation-level cloud certificate (for example AWS Cloud Practitioner or Azure AZ-900)
Year 2	Data structures and algorithms, databases, web or backend development, and networking basics	Build a full project and join at least one hackathon; start learning cyber security fundamentals
Year 3	Cloud services in depth, security practices, and team-based software development	Complete an internship; earn an associate-level cloud or security certification
Year 4	Advanced projects, system design basics, and placement preparation	Finish a final-year capstone project and secure a software engineering placement

5. Key Takeaways

- **Strong base first:** a solid grasp of C, C++, and Python and of core concepts will support every advanced skill I add later.
- **Turn weaknesses into habits:** planning my time and practising public speaking regularly will make me a more complete professional.
- **Learn by building:** projects, hackathons, and internships will give me the experience that I lack today.

6. Presentation Script (Speaking Notes)

Slide 1: Introduction and Vision

> *"Good morning, sir and classmates. My name is Haranya I, a first-year Computer Science and Engineering student at St. Joseph's College of Engineering. Today I will present my SWOC analysis for Professional English. My goal is to become a software engineer and developer with strong skills in cloud computing and cyber security."*

Slide 2: Strengths and Weaknesses

> *"My main strengths are that I learn quickly, work well in a team, and focus on truly understanding concepts. I also know C, C++, and Python. My weaknesses are managing my time and speaking confidently in front of people, and I am working on both."*

Slide 3: Opportunities and Challenges

> *"Cloud computing and cyber security are growing fields, and there are many student programs, certifications, and hackathons that I can use. The challenges are that technology changes quickly and I do not have an internship yet, so I plan to keep learning steadily and look for practical experience."*

Slide 4: Plan and Conclusion

> *"My four-year roadmap moves from programming fundamentals in the first year to an internship in the third year and a strong final project and placement in the fourth. By building on my strengths and improving my weaknesses step by step, I am confident of reaching my goal. Thank you, and I welcome your questions."*

Integrated Learning Report: Soft Skill Development

Name: Haranya I
College Name: St. Joseph's College of Engineering
Course Name: B.E. Computer Science and Engineering (CSE)
NPTEL Course: Online Certification Course
Instructors: Prof. Priyadarshi Patnaik, Prof. V. N. Giri, Prof. D. Suar (IIT Kharagpur)
Duration: 8 Weeks

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1. Executive Summary

The "Soft Skill Development" course provides a comprehensive framework for enhancing interpersonal and intrapersonal competencies essential for professional success. Unlike technical training, this course focuses on the "how" of communication and behavior. It progresses from foundational communication mechanics (listening, speaking, non-verbal cues) to complex interpersonal dynamics (group behavior, emotional intelligence) and finally to self-management traits (creativity, leadership, stress management). The core objective is to transform students from passive learners into effective communicators who can navigate workplace challenges, resolve conflicts, and lead teams with emotional maturity.

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2. Integrated Week-by-Week Learning Diary

Week 1: Communication Skills – The Basics

- **Key Topics:** Understanding the communicative environment, active listening, when to speak, starting conversations.
- **Learning Insight:** I learned that communication is not just about speaking but heavily relies on listening. The concept of "what to listen for" helped me realize I often listen to reply rather than to understand.
- **Personal Reflection:** *[Insert a specific instance where you practiced active listening this week, e.g., in a group discussion or with a friend.]*

Week 2: Presentation and Interaction

- **Key Topics:** Multimedia presentation basics, communication styles, speaking in groups.
- **Learning Insight:** The distinction between different communication styles (passive, aggressive, assertive) was eye-opening. I realized the importance of being assertive without being aggressive.
- **Personal Reflection:** *[Describe how you prepared for a presentation or spoke up in a group setting using these techniques.]*

Week 3: Visual, Nonverbal, and Aural Communication

- **Key Topics:** Visual culture, body language, facial expressions, the impact of tone (aural).
- **Learning Insight:** A significant portion of communication is non-verbal. I became more aware of my own body language and how facial expressions can contradict spoken words.
- **Personal Reflection:** *[Mention an observation where you noticed someone's body language telling a different story than their words.]*

Week 4: Interpersonal Communication – Individuals and Groups

- **Key Topics:** Building relationships, group dynamics, conflict resolution, social media identity.
- **Learning Insight:** Understanding "Group Dynamics" explained why some teams fail while others succeed. The module on conflict resolution provided practical steps to de-escalate tension.
- **Personal Reflection:** *[Reflect on a team project or group activity where you applied conflict resolution strategies.]*

Week 5: Emotional and Social Skills

- **Key Topics:** Emotional intelligence (EQ), empathy, intrapersonal communication, change tolerance.
- **Learning Insight:** EQ is as critical as IQ. The ability to empathize and manage one's own emotions during stress is a skill I need to develop further.
- **Personal Reflection:** *[Describe a situation where you managed your emotions effectively or showed empathy.]*

Week 6: Developing Key Traits – Creativity and Problem Solving

- **Key Topics:** Creativity definition, critical thinking, problem-solving frameworks.
- **Learning Insight:** Creativity is not just artistic; it is a problem-solving tool. The course taught methods to approach problems from multiple angles (lateral thinking).
- **Personal Reflection:** *[Share how you approached an academic or personal problem differently this week.]*

Week 7: Motivation, Persuasion, and Leadership

- **Key Topics:** Self-motivation, the art of persuasion, negotiation, leadership qualities.
- **Learning Insight:** Persuasion is about understanding the other party's needs, not just arguing. Leadership is about motivating others, not just commanding them.
- **Personal Reflection:** *[Discuss a time you tried to persuade someone or took a leadership role.]*

Week 8: Essential Vocational Skills – Survival Strategies

- **Key Topics:** Time management, stress management, resilience, work-life balance, applying skills to the workplace.
- **Learning Insight:** The "Work-Life Balance" module was crucial. I learned that resilience is about bouncing back from failure, not avoiding it.
- **Personal Reflection:** *[Detail a time-management technique you adopted, such as the Pomodoro technique or prioritization matrix.]*

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3. Conclusion

The NPTEL "Soft Skill Development" course has been a transformative journey from being a technically competent student to a well-rounded professional candidate. The progression from basic communication mechanics to advanced emotional intelligence and leadership has equipped me with the tools to handle real-world workplace scenarios. The most significant takeaway is that soft skills are not "innate talents" but "learned behaviors" that require consistent practice and self-awareness.

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4. Future Applicability

- **Job Interviews:** I will apply the "Presentation Skills" and "Assertive Communication" techniques to articulate my strengths confidently during interviews.
- **Team Projects:** The "Group Dynamics" and "Conflict Resolution" modules will be directly applicable to my final year project and future collaborative work environments.
- **Leadership Roles:** As I aim for managerial positions, the lessons on "Motivation" and "Emotional Intelligence" will help me lead teams effectively.
- **Personal Growth:** The "Stress Management" and "Work-Life Balance" strategies will ensure I maintain mental well-being while pursuing a demanding career in engineering/technology.

BIOGRAPHICAL PROFILE

Mark Zuckerberg

Co-Founder of Facebook | Chairman and CEO of Meta Platforms

1. Early Life and Education

Mark Elliot Zuckerberg was born on **May 14, 1984**, in White Plains, New York, and grew up in nearby Dobbs Ferry with his three sisters. His mother, Karen, worked as a psychiatrist and his father, Edward, as a dentist. Education and curiosity were valued at home, and Mark showed an early talent for programming. After a short time at Ardsley High School, he moved to **Phillips Exeter Academy**, where he continued to sharpen his technical and academic skills.

In 2002 he entered **Harvard University**, where he studied computer science and psychology and quickly built a reputation as a gifted programmer. Harvard became the place where his interest in code met his interest in how people connect with one another, and that combination shaped everything that followed.

2. Building Facebook: Key Milestones

In January 2004 Zuckerberg began writing the code for a new student directory. On **February 4, 2004**, he launched it as thefacebook.com, working with his Harvard roommates Eduardo Saverin, Andrew McCollum, Dustin Moskovitz, and Chris Hughes. It spread quickly to other universities, and in the summer of 2004 the team moved to Palo Alto, California. Zuckerberg left Harvard to focus on the company. The major steps in the years that followed are summarised below.

1984	Born on May 14 in White Plains, New York.
2002	Graduates from Phillips Exeter Academy and enters Harvard University.
2003	Builds Facemash, an early campus website that gets him noticed.
2004	Launches thefacebook.com on February 4 with his Harvard roommates; moves to Palo Alto in the summer; the site passes one million users by year-end.
2005	The site drops “the” and becomes simply Facebook.
2007	Microsoft buys a small stake in the company, giving it a major outside valuation.
2012	Facebook goes public and reaches one billion users; it also acquires Instagram.
2014	Acquires WhatsApp and Oculus VR.
2015	Launches the Chan Zuckerberg Initiative with his wife, Priscilla Chan.
2017	Receives an honorary degree from Harvard.
2021	The parent company is renamed Meta Platforms.

Along the way the company faced legal disputes over its origins and continuing public debate about privacy and its influence on society. These are part of the full story, and they show that building technology at a global scale brings serious responsibilities as well as success.

3. What Made Him Successful

Zuckerberg's rise was driven by a mix of technical ability, bold decisions, and an unusually long-term outlook. He wrote much of the early code himself, so he understood the product from the inside. He also kept control of the company's direction, which allowed him to make large bets, such as buying Instagram and WhatsApp, and later moving the company toward virtual reality and artificial intelligence.

- **Hands-on technical skill:** he began as a programmer and stayed close to the product as it grew.
- **Focus on connection:** he built around a simple human need, which is staying in touch with other people.
- **Willingness to take risks:** he left university, declined an early large buyout offer, and repeatedly invested in new platforms.
- **Adaptability:** he moved the company from desktop websites to mobile, and then toward immersive technology and AI.

4. Lessons for CSE Students

- **Build real things:** a working project teaches more than theory alone, so start building early and keep improving.
- **Master the fundamentals:** strong programming, data structures, and databases make large systems possible.
- **Think about scale:** ask how a solution would serve thousands or millions of users, not just one.
- **Stay curious about people:** good software solves human problems, so understanding users matters as much as code.
- **Accept responsibility:** technology affects society, so privacy, ethics, and trust must be part of every design.

5. Guiding Mottos Linked to His Work

Facebook's early engineering culture, often called “the Hacker Way,” was known for two mottos that were displayed around its offices:

- **“Move fast and break things.”:** a reminder to build and experiment quickly rather than wait for perfection. The company later softened this idea as its products matured.
- **“Done is better than perfect.”:** a push to release, learn from users, and improve, instead of delaying a project indefinitely.

Today, Zuckerberg continues to lead Meta as chairman and CEO, with a strong focus on artificial intelligence and immersive technologies.

PROJECT PROPOSAL AND TECHNICAL ABSTRACT

Smart Campus Power Saver: Occupancy-Based Automatic Electricity Control

Candidate	Haranya I
Institution	St. Joseph's College of Engineering
Department	Computer Science and Engineering (CSE)
Academic Year	First Year, B.E. CSE
Course Subject	Professional English

Technical Abstract

Classrooms, laboratories, and study halls on a campus often keep their lights and fans running long after the last person has left, which adds avoidable cost to electricity bills and unnecessary load on the power grid. This project proposes the **Smart Campus Power Saver**, a closed-loop system that switches off electrical appliances automatically when a room is empty. A standard camera feed is analysed by a **Python program using the OpenCV library**, which compares consecutive video frames to detect movement and decide whether the room is occupied. When no movement is found for a continuous period of five minutes, the program sends a shutdown command through the PySerial library to an **Arduino Uno microcontroller**. The Arduino then drives a **5V relay module** that disconnects the lights and fans of that room. The prototype is low in cost, reuses existing cameras, and is estimated to reduce idle energy use in classrooms by up to 35%, a figure that will be confirmed through testing.

1. Problem Statement and Background

Educational campuses consume large amounts of electricity every day, and a considerable share is wasted on lights, ceiling fans, and air conditioners that stay on in unoccupied rooms. Switching them off currently depends on individuals remembering to do so, and there is no central way to monitor or correct the lapses. The result is higher energy bills, extra strain on the electrical supply, and a larger environmental footprint for the institution.

2. Proposed Methodology

A. Image Acquisition

A webcam, or an existing classroom camera, continuously supplies live video of the room to a computer that runs the monitoring program.

B. Motion Detection with Python and OpenCV

The program converts each frame to grayscale, smooths it to remove noise, and compares it with the previous frame and a running background image. Differences above a set threshold are treated as genuine movement, while small changes such as lighting flicker are ignored.

C. Decision Logic and Timer

The program keeps track of the time since movement was last seen. If the room stays inactive for a full five minutes, it concludes that the room is vacant and prepares a shutdown command. Any new movement resets the timer immediately.

D. Relay Control through Arduino

The command travels over a USB serial connection to the Arduino Uno, which changes the state of its output pins. These pins trigger the relay module, which cuts the power supply to the room's lights and fans. When movement returns, the same path can restore the supply.

3. Technology Stack

Programming Language	Python 3.x
Computer Vision Library	OpenCV
Input Device	Webcam or existing classroom camera
Microcontroller	Arduino Uno
Switching Module	5V multi-channel relay module with optocoupler isolation
Communication Link	PySerial over a USB serial connection

4. Seminar Presentation Script

Slide 1: The Problem

> *"Good morning, sir and classmates. Every day our campus loses electricity in rooms that nobody is using. Lights and fans stay on in empty laboratories and halls simply because people forget to switch them off."*

Slide 2: Our Solution

> *"To address this, I propose the Smart Campus Power Saver. Instead of installing costly sensors in every room, it uses a camera that is already available, together with a Python program that works out whether anyone is present."*

Slide 3: How It Works

> *"The program compares each video frame with the one before it to detect movement. If a room shows no movement for five minutes, it sends a signal through the serial port to an Arduino, which activates a relay and cuts power to the lights and fans."*

Slide 4: Expected Outcome

> *"This design offers a low-cost, practical way to reduce wasted energy, with an estimated saving of up to 35%. It shows how simple, well-structured code can solve a real problem on our own campus. Thank you, and I welcome your questions."*

5. Expected Outcomes and Institutional Impact

- **Practical application of programming:** the project turns basic Python and electronics knowledge into a working solution to a real energy problem.
- **Progress towards a green campus:** automatic power saving can help the college reduce its electricity use and support its sustainability goals.
- **Early engineering skills:** combining software and hardware in one system builds problem-solving ability and confidence from the first year.
- **Scope for improvement:** future versions can use person detection so that people sitting still are not mistaken for an empty room, and can add scheduling, usage dashboards, and predictive models.
- **Safety consideration:** relays must be rated for the electrical load, and all mains wiring should be done or checked by a qualified person.

>> CREDITS • ONLINE PROFILES

GitHub, LinkedIn & Practical Learning

Showcasing my skills, projects and professional presence

My GitHub profile reflects my coding journey in C, C++ and Python, while my LinkedIn profile highlights my academic background, skills and interests. I also actively learn through online platforms like NPTEL, Coursera and other resources to stay updated with the latest technologies.

GitHub Profile

Username : **Haranya-I**

Link : **github.com/Haranya-I**

Includes my coding projects like Food Waste Tracker, basic C programs and Python projects.

LinkedIn Profile

Name : **Haranya.I**

Profile Link : **linkedin.com/in/haranya-i**

Connect with me for academic collaborations and opportunities.

Online Learning

Platforms : NPTEL, Coursera, Udemy

Courses : Python for Data Science, Cybersecurity Basics, Cloud Computing

Practical Approach

Focus : Hands-on learning, building real-time projects and improving continuously.

PROFILE LINKS

- GitHub : https://github.com/Haranya-I
- LinkedIn : https://www.linkedin.com/in/haranya-i
- LeetCode : https://leetcode.com/u/Haranya_I
- Portfolio : https://haranya-portfolio.vercel.app

PROFILE IMPROVEMENT CHECKLIST

Use a clear headline, a concise about section, showcase my projects, and add relevant skills and certifications. I regularly update my GitHub with new projects and complete online courses to strengthen my profile for placements and higher studies.

Take Reading Test | Your Past Results

Current Test Results

READING SPEED

272

Words per Minute (WPM)

COMPREHENSION

75%

of questions answered correctly

At your current pace, a 10-minute article takes about **9 minutes**. With SwiftRead's up to 3X faster reading, you could save about **18 minutes/day** (assuming 3 similar articles per day).

Take another test

TRAIN FASTER WITH SWIFTREAD

Take Reading Test | Your Past Results

Current Test Results

READING SPEED

228

Words per Minute (WPM)

COMPREHENSION

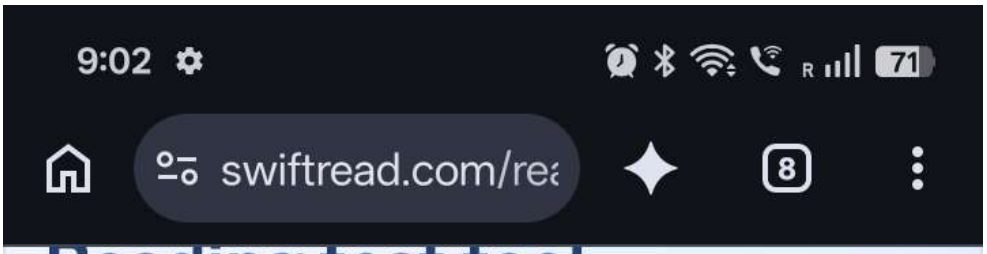
50%

of questions answered correctly

At your current pace, a 10-minute article takes about **11 minutes**. With SwiftRead's up to 3X faster reading, you could save about **22 minutes/day** (assuming 3 similar articles per day).

Take another test

TRAIN FASTER WITH SWIFTREAD



Reading test tool

Takes about 1 minute and gives instant speed + comprehension feedback.

1 Start

2 Read

3 Results

Take Reading Test | Your Past Results

Current Test Results

READING SPEED

245

Words per Minute (WPM)

COMPREHENSION

33%

of questions answered correctly